

Minimal Notes

Fourier Analysis

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WEEK 1 - BASIC DEFINITIONS

Fourier Analysis is roughly concerned with the following question: Given a function $f : [-\pi, \pi] \rightarrow \mathbb{R}$, when can we write f as a sum of trigonometric polynomials e^{inx} with $n \in \mathbb{Z}$ so that $f(x) = \sum_{n \in \mathbb{Z}} a_n e^{inx}$?

In the following function means periodic function (until week 6).

Definition 1 (Fourier Coefficients and Series). Given $f : [a, b] \rightarrow \mathbb{R}$ Riemann Integrable (RI) we define the n -th Fourier coefficient (FC)

$$\hat{f}(n) = \frac{1}{b-a} \int_a^b f(x) e^{-2\pi i n x / L} dx$$

and the corresponding Fourier series (FS)

$$\sum_{n \in \mathbb{Z}} \hat{f}(n) e^{2\pi i n x / L},$$

with partial sum denoted

$$S_N(f)(x) = \sum_{n=-N}^N \hat{f}(n) e^{2\pi i n x / L}.$$

Example 1: • (Dirichlet Kernel) $D_N(x) = \sum_{n=-N}^N e^{inx} = \frac{\sin((N+\frac{1}{2})x)}{\sin(x/2)}$

• (Poisson Kernel) $P_r(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} e^{in\theta} = \frac{1-r^2}{1-2r \cos \theta + r^2}$ with $0 \leq r < 1$ and $\theta \in [-\pi, \pi]$.

Proofs. • (Closed form of Dirichlet Kernel)

$$\begin{aligned} D_N(x) &= \sum_{k=-N}^N e^{ikx} \\ &= e^{-iNx} + e^{-i(N-1)x} + \dots + e^{i(N-1)x} + e^{iNx} \\ &= e^{-iNx} (1 + e^{ix} + \dots + e^{2iNx}) \\ &= e^{-iNx} \cdot \frac{1 - e^{i(2N+1)x}}{1 - e^{ix}} \quad (\text{geometric series}) \\ &= \frac{e^{-iNx} - e^{i(N+1)x}}{1 - e^{ix}} \\ &= \frac{e^{-iNx} - e^{i(N+1)x}}{e^{-ix/2}(e^{ix/2} - e^{-ix/2})} \cdot \frac{e^{ix/2}}{e^{ix/2}} \\ &= \frac{e^{i(N+1/2)x} - e^{-i(N+1/2)x}}{e^{ix/2} - e^{-ix/2}} \\ &= \frac{2i \sin((N + \frac{1}{2})x)}{2i \sin(x/2)} \\ &= \frac{\sin((N + \frac{1}{2})x)}{\sin(x/2)}. \end{aligned}$$

- (Closed form of Fejer Kernel)

$$\begin{aligned}
F_N(x) &= \frac{1}{N} \sum_{k=0}^{N-1} D_k(x) \\
&= \frac{1}{N} \sum_{k=0}^{N-1} \frac{\sin((k + \frac{1}{2})x)}{\sin(x/2)} \\
&= \frac{1}{N \sin(x/2)} \sum_{k=0}^{N-1} \sin\left(\left(k + \frac{1}{2}\right)x\right) \\
&= \frac{1}{N \sin(x/2)} \cdot \frac{\sin^2(Nx/2)}{\sin(x/2)} \quad \left(\text{using } \sum_{k=0}^{N-1} \sin((k + \frac{1}{2})x) = \frac{\sin^2(Nx/2)}{\sin(x/2)}\right) \\
&= \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}.
\end{aligned}$$

Formula can be obtained again by using geometric series and the sin definition

$$\begin{aligned}
\sum_{k=0}^{N-1} \sin\left(\left(k + \frac{1}{2}\right)x\right) &= \text{Im} \sum_{k=0}^{N-1} e^{i(k+1/2)x} \\
&= \text{Im} \left(e^{ix/2} \sum_{k=0}^{N-1} e^{ikx} \right) \\
&= \text{Im} \left(e^{ix/2} \cdot \frac{1 - e^{iNx}}{1 - e^{ix}} \right) \quad (\text{geometric series}) \\
&= \text{Im} \left(e^{ix/2} \cdot \frac{e^{iNx/2}(e^{-iNx/2} - e^{iNx/2})}{e^{ix/2}(e^{-ix/2} - e^{ix/2})} \right) \\
&= \text{Im} \left(\frac{e^{iNx/2} \cdot (-2i \sin(Nx/2))}{-2i \sin(x/2)} \right) \\
&= \text{Im} \left(e^{iNx/2} \cdot \frac{\sin(Nx/2)}{\sin(x/2)} \right) \\
&= \frac{\sin(Nx/2)}{\sin(x/2)} \cdot \text{Im} e^{iNx/2} \\
&= \frac{\sin(Nx/2)}{\sin(x/2)} \cdot \sin(Nx/2) \\
&= \frac{\sin^2(Nx/2)}{\sin(x/2)}. \quad \square
\end{aligned}$$

WEEK 2 - UNIQUENESS OF FS, CONVOLUTIONS, KERNELS

Theorem 1 (Uniqueness of FS). • (Thm) f is RI with $\hat{f}(n) = 0$ for all $n \in \mathbb{Z} \implies f(\theta) = 0$ at every point θ where f is cts.

- (Corr) If f cts and $\hat{f}(n) = 0$ for all $n \in \mathbb{Z} \implies f = 0$.
- (Corr) Suppose f is cts and $\sum_{n \in \mathbb{Z}} |\hat{f}(n)| < \infty$. Then FS converges uniformly to f .

Definition 2 (Convolution). Suppose f, g are RI on $[-\pi, \pi]$ then $f * g$ is defined on $[-\pi, \pi]$ as

$$(f * g)(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y)g(x-y) dy$$

It has the following properties

- $f * (g + h) = (f * g) + (f * h)$
- $f * g$ cts
- $c(f * g) = (cf) * g = f * (cg)$
- $\widehat{f * g}(n) = \hat{f}(n)\hat{g}(n)$.
- $f * g = g * f$
- $\hat{f}'(n) = in\hat{f}(n)$
- $f * (g * h) = (f * g) * h$

Proof of last formulas:

$$\begin{aligned} \widehat{f * g}(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} (f * g)(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1}{2\pi} \left(\int_{-\pi}^{\pi} f(y)g(x-y) dy \right) e^{-inx} dx \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} g(x-y) e^{-in(x-y)} dx \right) dy \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(y) e^{-iny} \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} g(x) e^{-inx} dx \right) dy \\ &= \hat{f}(n)\hat{g}(n). \end{aligned}$$

and

$$\begin{aligned} \hat{f}'(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f'(x) e^{-inx} dx \\ &= \frac{1}{2\pi} \left[f(x) e^{-inx} \Big|_{-\pi}^{\pi} - \int_{-\pi}^{\pi} f(x) (-in) e^{-inx} dx \right] \quad (\text{By Parts}) \\ &= \frac{1}{2\pi} (f(\pi) e^{-in\pi} - f(-\pi) e^{in\pi}) + \frac{in}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx \\ &= 0 + in \cdot \hat{f}(n) \quad (\text{periodicity}) \\ &= in \hat{f}(n). \end{aligned}$$

□

We can use convolution to write (purely from definition of FC, check yourself!)

$$S_N(f)(x) = (f * D_N)(x)$$

Definition 3 (Good Kernel). Periodic functions $K_n(x)$ on $[-\pi, \pi]$ are family of good kernels if

- (Normalized) $\frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(x) dx = 1$.
- (Bounded) There is $M > 0$ with $\frac{1}{2\pi} \int_{-\pi}^{\pi} |K_n(x)| dx \leq M$ for all n .
- (Concentration near origin) For every $\delta > 0$ $\frac{1}{2\pi} \int_{\delta \leq |x| \leq \pi} |K_n(x)| dx \rightarrow 0$ as $n \rightarrow \infty$.

The following theorem says roughly that good kernels provide an approximation to nice functions at a point. Namely good kernels essentially approximate the Dirac delta function which uses integration to pick up a value of a function at a single point.

Theorem 2 (Approximation to Identity aka Good Kernels Thm). K_n family of good kernels, $f \in RI$. Then

$$\lim_{n \rightarrow \infty} (f * K_n)(x) = f(x) \quad \text{whenever } f \text{ cts at } x.$$

If f cts then the limit is uniform.

Proof. Idea: Divide and conquer. One part using continuity and the other using concentration property of good kernels.

From continuity: given $\varepsilon > 0$ there is δ so that $|y| < \delta$ so that $|f(x - y) - f(x)| < \varepsilon$. Then, by the first property of good kernels, we can write

$$\begin{aligned} |(f * K_n)(x) - f(x)| &= \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y) f(x - y) dy - f(x) \right| \\ &= \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y) f(x - y) dy - \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y) f(x) dy \right| \quad \left(\text{since } \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y) dy = 1 \right) \\ &= \left| \frac{1}{2\pi} \int_{-\pi}^{\pi} K_n(y) [f(x - y) - f(x)] dy \right| \\ &\leq \frac{1}{2\pi} \int_{|y| < \delta} |K_n(y)| |f(x - y) - f(x)| dy \\ &\quad + \frac{1}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| |f(x - y) - f(x)| dy \\ &\leq \frac{\varepsilon}{2\pi} \int_{-\pi}^{\pi} |K_n(y)| dy + \frac{2B}{2\pi} \int_{\delta \leq |y| \leq \pi} |K_n(y)| dy, \end{aligned}$$

where B is a bound for f (cts function on closed bounded interval is bounded). The first term is bounded by $\varepsilon M/2\pi$ because of the second property of good kernels. By the third property we see that for all large n , the second term will be less than ε . Therefore, for some constant $C > 0$ and all large n we have

$$|(f * K_n)(x) - f(x)| \leq C\varepsilon,$$

thereby proving the first assertion in the theorem. If f is continuous everywhere, then it is uniformly continuous, and δ can be chosen independent of x . This provides the desired conclusion that $f * K_n \rightarrow f$ uniformly. $\square$

WEEK 3 - SUMMABILITY

We can look at 'average of partial sums' version aka **Cesaro mean** of the Fourier series

$$\sigma_N(f) = \frac{S_0(f)(x) + \cdots + S_{N-1}(f)(x)}{N}.$$

We can write $\sigma_N(f)$ as convolution of f with so called **Fejér kernel** F_N defined as

$$F_N(x) = \frac{D_0(x) + \cdots + D_{N-1}(x)}{N} = \frac{1}{N} \frac{\sin^2(Nx/2)}{\sin^2(x/2)}.$$

Then

$$\sigma_N(f)(x) = (f * F_N)(x)$$

Applying the Approximation to Identity theorem we obtain

Theorem 3. • (Fejer's Thm) If f RI $\implies$ FS is Cesaro summable to f at every point where f is cts. If f cts everywhere then the convergence is uniform.

- (Corr) If f RI and $\hat{f}(n) = 0$ for all n then $f = 0$ where ever f is cts.
- (Weirstrass Approximation on Circle) Cts functions (on the circle ie periodic) can be uniformly approximated by trigonometric polynomials.

Another type of summability is **Abel summability** which uses a kind of geometric weight. If $f \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$ we define

$$A_r(f)(\theta) = \sum_{n=-\infty}^{\infty} r^{|n|} a_n e^{in\theta}.$$

Here in turn we can use the Poisson Kernel to write $A_r(f)(\theta) = (f * P_r)(\theta)$. Now using Approximation to identity again we obtain

Theorem 4. If f is RI then FS of f is Abel summable at every point of continuity. If f cts everywhere then the convergence is uniform.

Note. Standard summable $\implies$ Cesaro summable $\implies$ Abel summable. $\diamond$

Proofs. • (Convergent implies Cesaro convergent) Let $a_n \rightarrow L$. Define $\sigma_N = \frac{1}{N} \sum_{n=1}^N a_n$. Then

$$\begin{aligned} |\sigma_N - L| &= \left| \frac{1}{N} \sum_{n=1}^N (a_n - L) \right| \\ &\leq \frac{1}{N} \sum_{n=1}^K |a_n - L| + \frac{1}{N} \sum_{n=K+1}^N |a_n - L| \\ &< \frac{C}{N} + \frac{N-K}{N} \cdot \frac{\varepsilon}{2} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

for N large enough so that $\frac{C}{N} < \frac{\varepsilon}{2}$ and choosing K so that $|a_n - L| < \frac{\varepsilon}{2}$ for all $n > K$.

Therefore $\sigma_N \rightarrow L$. □

WEEK 4 - INNER PRODUCT

HILBERT SPACES AND COMPLETENESS

Recall that inner products $(\cdot, \cdot)$ satisfy

- (Cauchy-Schwarz) $|(f, g)| \leq \|f\| \|g\|$.
- (Norm) $\|f\|^2 = (f, f)$.
- (Triangle) $\|f + g\| \leq \|f\| + \|g\|$.

For RI functions f, g on $[0, 2\pi]$ the inner product is defined (note the conjugation on g !)

$$(f, g) = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \overline{g(\theta)} d\theta \quad \text{so} \quad \|f\|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta$$

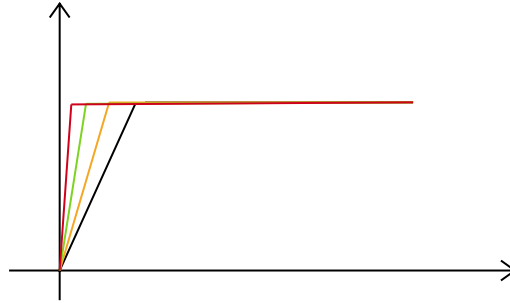
A **Hilbert space** is complete space with inner product. **Complete** means every Cauchy sequence is convergent to an element of the space. The space of RI function is not complete!

Example 2 (RI not complete): We need to find Cauchy sequence of RI function without limit. Take $f_n = 0$ for $x \leq \pi/n$ and $1/x^{1/4}$ for $x > \pi/n$. It can be seen this is a Cauchy sequence (directly from definition) but the limit is $1/x^{1/4}$ which is not integrable near zero so not RI. So even though we identified a limit, the limit is not in the space.

The completion of the space of RI functions using the above inner product is the Lebesgue space $L^2([0, 2\pi])$ with norm denoted $\|\cdot\|_2$. In this space the trigonometric polynomials form an **orthonormal set** ie $(e^{in\theta}, e^{im\theta}) = \delta_{nm}$. Notably

$$(f, e^{inx}) = \hat{f}(n).$$

Example 3 (Cts RI not complete): Picture example:



An example of a complete space is

Example 4 ($\ell^2(\mathbb{Z})$ is complete): Idea: show that each coordinate sequence is Cauchy so convergent because $\mathbb{R}$ is complete. Construct the limit from the coordinate limits and show the original sequence converges to it.

Let $\{x^{(n)}\}_{n=1}^\infty$ be a Cauchy sequence in $\ell^2(\mathbb{Z})$, where $x^{(n)} = (x_k^{(n)})_{k \in \mathbb{Z}}$ so

$$\forall \varepsilon > 0, \exists N \forall m, n \geq N : \|x^{(m)} - x^{(n)}\|_2^2 = \sum_{k \in \mathbb{Z}} |x_k^{(m)} - x_k^{(n)}|^2 < \varepsilon^2.$$

For each fixed k , $|x_k^{(m)} - x_k^{(n)}| \leq \|x^{(m)} - x^{(n)}\|_2 < \varepsilon$, so $\{x_k^{(n)}\}_n$ is Cauchy in $\mathbb{C}$. Hence $x_k^{(n)} \rightarrow x_k$ as $n \rightarrow \infty$ for each k . Define $x = (x_k)_{k \in \mathbb{Z}}$.

Now show $x \in \ell^2$ and $x^{(n)} \rightarrow x$ in ℓ^2 . For any K and $m, n \geq N$,

$$\sum_{|k| \leq K} |x_k^{(m)} - x_k^{(n)}|^2 \leq \|x^{(m)} - x^{(n)}\|_2^2 < \varepsilon^2.$$

Let $n \rightarrow \infty$ (keeping m fixed) to obtain

$$\sum_{|k| \leq K} |x_k^{(m)} - x_k|^2 \leq \varepsilon^2.$$

Since K is arbitrary, let $K \rightarrow \infty$:

$$\|x^{(m)} - x\|_2^2 = \sum_{k \in \mathbb{Z}} |x_k^{(m)} - x_k|^2 \leq \varepsilon^2 \quad \text{for all } m \geq N.$$

Thus $x^{(m)} \rightarrow x$ in ℓ^2 . Finally,

$$\|x\|_2 \leq \|x^{(m)}\|_2 + \|x^{(m)} - x\|_2 < \infty,$$

so $x \in \ell^2(\mathbb{Z})$. Hence every Cauchy sequence converges, and $\ell^2(\mathbb{Z})$ is complete.

APPROXIMATIONS AND FORMULAS IN HILBERT SPACES

Lemma 1 (Minimising Property). *If f is integrable on the circle with Fourier coefficients a_n , then*

$$\|f - S_N(f)\| \leq \left\| f - \sum_{|n| \leq N} c_n e_n \right\|$$

for any complex numbers c_n . Moreover, equality holds precisely when $c_n = a_n$ for all $|n| \leq N$.

Proof. This follows immediately by applying the Pythagorean theorem to

$$f - \sum_{|n| \leq N} c_n e_n = f - S_N(f) + \sum_{|n| \leq N} b_n e_n \quad (\text{where } b_n = a_n - c_n).$$

Since $f - S_N(f)$ is orthogonal to all e_n with $|n| \leq N$, the Pythagorean theorem gives

$$\begin{aligned} \left\| f - \sum_{|n| \leq N} c_n e_n \right\|^2 &= \|f - S_N(f)\|^2 + \sum_{|n| \leq N} |b_n|^2 \\ &\geq \|f - S_N(f)\|^2, \end{aligned}$$

with equality iff $b_n = 0$ for all $|n| \leq N$, i.e., $c_n = a_n$. □

An important equality is the **Parseval's identity** which is just generalization of the Pythagoras theorem. If $f \sim \sum_{n \in \mathbb{Z}} \hat{f}(n) e^{in\theta}$

$$\sum_{n=-\infty}^{\infty} |\hat{f}(n)|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 d\theta.$$

More generally for any orthonormal set e_n we have for $a_n = (f, e_n)$ **Bessel's inequality**

$$\sum_{n=-\infty}^{\infty} |a_n|^2 \leq \|f\|^2.$$

Central result of this week says roughly that the FS of RI functions converges in L^2 -norm to f :

Theorem 5 (L^2 convergence aka mean square convergence). *Let f be RI and $\sum a_n e^{in\theta}$ the corresponding FS. Then*

$$\|f - S_N f\|_2^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

From Bessel's inequality deduce

Theorem 6 (Riemann-Lebesgue lemma). $f \text{ RI} \implies \hat{f}(n) \rightarrow 0 \text{ as } n \rightarrow \infty$.

Theorem 7 (Differentiable implies Pointwise convergence). *Let f be RI and differentiable at θ_0 . Then*

$$S_N(f)(\theta_0) \rightarrow f(\theta_0) \quad \text{as } N \rightarrow \infty.$$

WEEK 5 - ISOPERIMETRIC INEQUALITY, WEYL EQUIDISTRIBUTION

ISOPERIMETRY

A famous question in mathematics is the so called isoperimetric problem: Given a fixed perimeter l , what curve encloses the largest area. The answer is a circle.

A curve in $\mathbb{R}^2$ can be parametrized as $\gamma(t) = (x(t), y(t))$ with $t \in [a, b]$. Its **length** is defined as

$$l(\gamma) = \int_a^b |\gamma'(t)| dt = \int_a^b \sqrt{x^2(t) + y^2(t)} dt.$$

The Area enclosed by a curve γ is

$$A(\gamma) = \frac{1}{2} \left| \int_a^b x(t)y'(t) - y(t)x'(t) ds \right|.$$

Theorem 8 (Isoperimetric Answer). *Suppose γ is non-self-intersecting (aka simple), closed (ends connected), curve in $\mathbb{R}^2$ of length l and area A . Then*

$$A \leq \frac{l^2}{4\pi}$$

with equality iff γ is a circle.

WEYL'S EQUIDISTRIBUTION THEOREM

Denote by $\langle x \rangle$ the fractional part of $x \in \mathbb{R}$. Informally a sequence ξ_n is equidistributed on $[0, 1)$ if there is no part of the interval contains greater concentration of the members of the sequence than any other part. Formally $\xi_n \in [0, 1)$ **equidistributed** if for any $(a, b) \subset [0, 1)$ we have

$$\lim_{N \rightarrow \infty} \frac{|\{1 \leq n \leq N : \xi_n \in (a, b)\}|}{N} = b - a.$$

This can be imagined similarly as a Monte Carlo approximation of an area ie the number of members of the sequence that fall in the interval in the long run is equal to the size of the interval.

Theorem 9 (Weyl Equidistribution). *Take γ irrational. The sequence of fractional parts $\langle \gamma \rangle, \langle 2\gamma \rangle, \langle 3\gamma \rangle, \dots$ is equidistributed in $[0, 1)$.*

More generally there is a criterion for deciding whether a sequence is equidistributed:

Theorem 10 (Weyl's criterion). *A sequence $\xi_n \in [0, 1)$ is equidistributed $\iff$ for all $k \neq 0$ we have*

$$\frac{1}{N} \sum_{n=1}^N e^{2\pi i k \xi_n} \rightarrow 0, \quad \text{as } N \rightarrow \infty.$$

WEEK 6 - FOURIER TRANSFORM

The Fourier Transform (FT) is roughly speaking a continuous analog of the FS. We say f is of **moderate decrease** $\mathcal{M}(\mathbb{R})$ if it is cts and there is $A > 0$ with

$$|f(x)| \leq \frac{A}{1+x^2} \quad \text{for all } x \in \mathbb{R}.$$

We can integrate such functions on the whole line by defining

$$\int_{-\infty}^{\infty} f(x) dx = \lim_{N \rightarrow \infty} \int_{-N}^N f(x) dx.$$

Such integral retains linearity, it is translation and scale invariant, and is continuous:

$$\int_{-\infty}^{\infty} |f(x-h) - f(x)| dx \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Definition 4 (Fourier Transform). Let $f \in \mathcal{M}(\mathbb{R})$ we define its FT for $\xi \in \mathbb{R}$ as

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx.$$

We want the FT itself to be well-behaved and require a more stringent space of functions to guarantee that. One such space where the FT definition makes sense and has good decay is the space of rapidly decreasing functions:

Definition 5 (Schwarz space). The Schwarz space $\mathcal{S}(\mathbb{R})$ is the space of C^∞ functions such that

$$\sup_{x \in \mathbb{R}} |x|^k |f^{(l)}(x)| < \infty \quad \text{for every } k, l \geq 0.$$

We collect some basic properties

Lemma 2.

- $\widehat{f(x+h)} = \hat{f}(\xi) e^{2\pi i h \xi}$
- (Schwarz formula 1) $\widehat{f'(x)} = 2\pi i \xi \hat{f}(\xi)$
- $\widehat{(f(x) e^{-2\pi i x h})} = \hat{f}(\xi + h)$
- $\widehat{f(\delta x)} = \delta^{-1} \hat{f}(\delta^{-1} \xi)$
- (Schwarz formula 2) $\widehat{(-2\pi i x f(x))} = \frac{d}{d\xi} \hat{f}(\xi).$

Notable example of a Schwarz function is the following.

Example 5 (Gaussian): The function $f(x) = e^{-\pi x^2}$ is called **Gaussian** and belongs to the Schwarz space. It turns out that $\hat{f}(\xi) = f(\xi)$. Indeed Define

$$F(\xi) = \hat{f}(\xi) = \int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x \xi} dx,$$

and observe that $F(0) = 1$. By the second Schwarz formula, and the fact that $f'(x) = -2\pi x f(x)$, we obtain

$$\begin{aligned} F'(\xi) &= \int_{-\infty}^{\infty} f(x) (-2\pi i x) e^{-2\pi i x \xi} dx && \text{(differentiating under the integral)} \\ &= i \int_{-\infty}^{\infty} f'(x) e^{-2\pi i x \xi} dx && \text{(since } f'(x) = -2\pi x f(x)). \end{aligned}$$

By the first Schwarz formula, we find that

$$\begin{aligned} F'(\xi) &= i(2\pi i\xi)\hat{f}(\xi) \quad (\text{by } \widehat{f'}(\xi) = 2\pi i\xi\hat{f}(\xi)) \\ &= -2\pi\xi F(\xi). \end{aligned}$$

If we define $G(\xi) = F(\xi)e^{\pi\xi^2}$, then from what we have seen above, it follows that $G'(\xi) = 0$, hence G is constant. Since $F(0) = 1$, we conclude that G is identically equal to 1, therefore $F(\xi) = e^{-\pi\xi^2}$, as was to be shown. $\square$

In general it is very difficult to describe the image of a space under the FT. This is where Schwarz space shines because the FT of a Schwarz function is again Schwarz.

Theorem 11 (FT Schwarz is Schwarz). $f \in \mathcal{S}(\mathbb{R}) \implies \hat{f} \in \mathcal{S}(\mathbb{R})$. Furthermore if $f, g \in \mathcal{S}(\mathbb{R})$ then $f * g \in \mathcal{S}(\mathbb{R})$, $f * g = g * f$ and $\widehat{f * g}(\xi) = \hat{f}(\xi)\hat{g}(\xi)$.

Theorem 12 (Fourier Inversion). Let $f, g \in \mathcal{S}(\mathbb{R})$

- $\int_{-\infty}^{\infty} f(x)\hat{g}(x) dx = \int_{-\infty}^{\infty} g(x)\hat{f}(x) dx$.
- $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i x \xi} d\xi$.

Proof. We show the latter formula. We first claim that

$$f(0) = \int_{-\infty}^{\infty} \hat{f}(\xi) d\xi.$$

Let $G_\delta(x) = e^{-\pi\delta x^2}$ so that $\hat{G}_\delta(\xi) = K_\delta(\xi)$. By the first formula we get

$$\int_{-\infty}^{\infty} f(x)K_\delta(x) dx = \int_{-\infty}^{\infty} \hat{f}(\xi)G_\delta(\xi) d\xi.$$

Since K_δ is a good kernel, the first integral goes to $f(0)$ as δ tends to 0. Since the second integral clearly converges to $\int_{-\infty}^{\infty} \hat{f}(\xi) d\xi$ as δ tends to 0, our claim is proved. In general, let $F(y) = f(y+x)$ so that

$$f(x) = F(0) = \int_{-\infty}^{\infty} \hat{F}(\xi) d\xi = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i x \xi} d\xi. \quad \square$$

The following theorem says FT preserves norms on the Schwarz space

Theorem 13 (Plancherel). If $f \in \mathcal{S}(\mathbb{R})$ then $\|f\|_2 = \|\hat{f}\|_2$.

Now we can formulate the full Weirstrass Approximation theorem which says polynomials are dense in $C([a, b])$.

Theorem 14 (Weirstrass Approximation). Let $f \in C([a, b])$. For any $\varepsilon > 0$ there exists a polynomials P so that

$$\|f - P\|_\infty := \sup_{x \in [a, b]} |f(x) - P(x)| < \varepsilon.$$

PDE EXAMPLES

The **heat equation** is

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad \text{with IC } u(x, 0) = f(x).$$

The solution is given using the heat kernel

$$H_t(x) = K_\delta(x) \quad \text{with } \delta = 4\pi t$$

so in particular

$$H_t(x) = \frac{1}{(4\pi t)^{1/2}} e^{-x^2/4t} \quad \text{and} \quad \hat{H}_t(\xi) = e^{-4\pi^2 t \xi^2}.$$

Theorem 15 (Heat equation solution). *Given $f \in \mathcal{S}(\mathbb{R})$ let*

$$u(x, t) = (f * H_t)(x) \quad t > 0.$$

Then

- $u \in C^2$ when $x \in \mathbb{R}, t > 0$ and solves the heat equation.
- $u(x, t) \rightarrow f(x)$ uniformly in x as $t \rightarrow 0$.
- $\int_{-\infty}^{\infty} |u(x, t) - f(x)|^2 dx \rightarrow 0$ as $t \rightarrow 0$.

Next we are concerned with the **Laplace equation**

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

on the upper half-plane subject to the boundary condition $u(x, 0) = f(x)$. The solution to this equation uses the Poisson kernel.

$$P_y(x) = \frac{y}{\pi(x^2 + y^2)} \quad \text{with } x \in \mathbb{R} \text{ and } y > 0.$$

Theorem 16 (Laplace equation solution). *Given $f \in \mathcal{S}(\mathbb{R})$ and let*

$$u(x, y) = (f * P_y)(x).$$

Then

- $u(x, y) \in C^2$ in upper half-plane and solves $\Delta u = 0$.
- $u(x, y) \rightarrow f(x)$ uniformly as $y \rightarrow 0$.
- $\int_{-\infty}^{\infty} |u(x, y) - f(x)|^2 dx \rightarrow 0$ as $y \rightarrow 0$.

Additionally we have uniqueness result

Theorem 17 (Laplace uniqueness). *Suppose u is cts on $\overline{\mathbb{R}_+^2}$, solves $\Delta u = 0$, $u(x, 0) = 0$ and vanishes at infinity. Then $u = 0$.*

Note. Note both H_t and P_y are good kernels. $\diamond$

WEEK 7 - POISSON SUMMATION, HEISENBERG UNCERTAINTY

Theorem 18 (Poisson Summation formula). *If $f \in \mathcal{S}(\mathbb{R})$, then*

$$\sum_{n=-\infty}^{\infty} f(x+n) = \sum_{n=-\infty}^{\infty} \hat{f}(n) e^{2\pi i x n}.$$

In particular, setting $x = 0$ we obtain

$$\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=-\infty}^{\infty} \hat{f}(n).$$

Note. Note both sides are a 1-periodic functions constructed from the f defined on $\mathbb{R}$. It is an essential connection between the discrete FS and its continuous analogue the FT. $\diamond$

Proof. To check the first formula it suffices, by **Theorem 1**, to show that both sides (which are continuous) have the same Fourier coefficients (viewed as functions on the circle). Clearly, the m -th Fourier coefficient of the right-hand side is $\hat{f}(m)$. For the left-hand side we have

$$\begin{aligned} \int_0^1 \left(\sum_{n=-\infty}^{\infty} f(x+n) \right) e^{-2\pi i m x} dx &= \sum_{n=-\infty}^{\infty} \int_0^1 f(x+n) e^{-2\pi i m x} dx \\ &= \sum_{n=-\infty}^{\infty} \int_n^{n+1} f(y) e^{-2\pi i m y} dy \quad (y = x+n) \\ &= \int_{-\infty}^{\infty} f(y) e^{-2\pi i m y} dy \\ &= \hat{f}(m). \end{aligned} \quad \square$$

Theorem 19 (Heisenberg Uncertainty). *Suppose $\psi \in \mathcal{S}(\mathbb{R})$ such that $\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$. Then*

$$\left(\int_{-\infty}^{\infty} x^2 |\psi(x)|^2 dx \right) \left(\int_{-\infty}^{\infty} \xi^2 |\hat{\psi}(\xi)|^2 d\xi \right) \geq \frac{1}{16\pi^2}$$

WEEK 8-11 - FOURIER TRANSFORM ON $\mathbb{R}^d$

Recall the multiindex notation with $\alpha = (\alpha_1, \dots, \alpha_d)$

$$\begin{aligned} x^\alpha &= x_1^{\alpha_1} \cdots x_d^{\alpha_d} \\ \left(\frac{\partial}{\partial x} \right)^\alpha &= \left(\frac{\partial}{\partial x_1} \right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_d} \right)^{\alpha_d} \end{aligned}$$

with $|\alpha| = \alpha_1 + \cdots + \alpha_d$. We can now define the Schwarz space on $\mathbb{R}^d$

Definition 6 ($\mathcal{S}(\mathbb{R}^d)$). *We have $f \in \mathcal{S}(\mathbb{R}^d)$ if $f \in C^\infty(\mathbb{R}^d)$ and for every multiindex α, β*

$$\sup_{x \in \mathbb{R}^d} |x^\alpha \left(\frac{\partial}{\partial x} \right)^\beta f(x)| < \infty.$$

Then we define the FT on $\mathcal{S}(\mathbb{R}^d)$ as

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \cdot \xi} dx.$$

with the only difference being the dot product between x and ξ . The same properties as for the 1-dimensional transform hold (see **Lemma 2**) with the exception of the important schwarz formulas

$$\left[\left(\frac{\partial}{\partial x} \right)^\alpha f(x) \right]^\wedge = (2\pi i \xi)^\alpha \hat{f}(\xi) \quad \text{and} \quad [(-2\pi i x)^\alpha f(x)]^\wedge = \left(\frac{\partial}{\partial x} \right)^\alpha \hat{f}(\xi)$$

Additionally if R is a rotation then we have

$$\widehat{f(Rx)} = \hat{f}(R\xi)$$

Fourier inversion and Plancherel still hold here:

$$f(x) = \int_{\mathbb{R}^d} \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi \quad \text{and} \quad \|f\|_2 = \|\hat{f}\|_2.$$

So FT remains isometrical isomorphism on $\mathcal{S}(\mathbb{R}^d)$. Similarly $(f * g)^\wedge = \hat{f}\hat{g}$.

ALBERT'S TIPS

- If you can answer all the questions on the revision sheet provided by Prof. Pushnitski then you should score 1st. (many answered here, others were exercises like even and odd relation etc.)
- The proofs and examples in these minimal notes are good to know for the exam.
- Try 1 or 2 past papers.
- If you have time have a look at the easier exercises from each weekly sheet.
- Send me an email if you need help.
- If a proof is easy or short it might be examinable (formulas especially common).